[INCH-POUND]

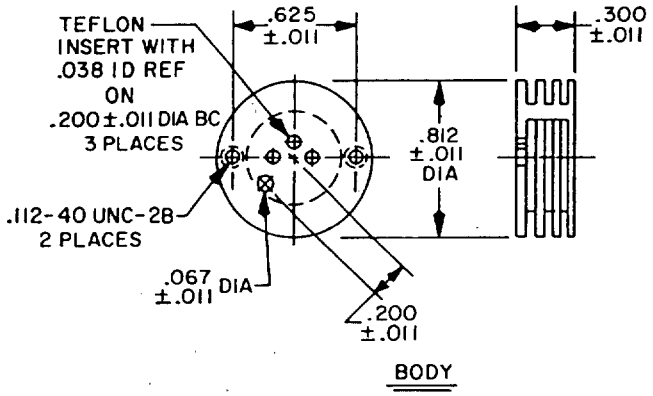
MIL-H-87111/4D
24 Oct 1991
SUPERSEDING
MIL-H-87111/4C
23 November 1990

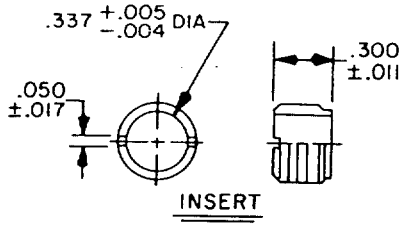
MILITARY SPECIFICATION SHEET

(D) HEAT SINK, SEMICONDUCTOR DEVICES, CLASS 1 (ENCAPSULATING TYPE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

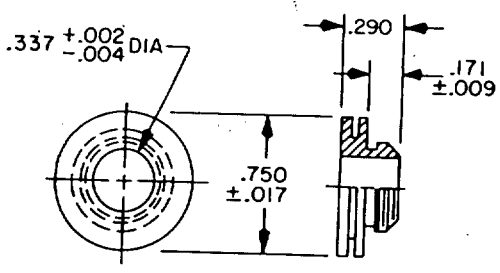
The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-H-87111.

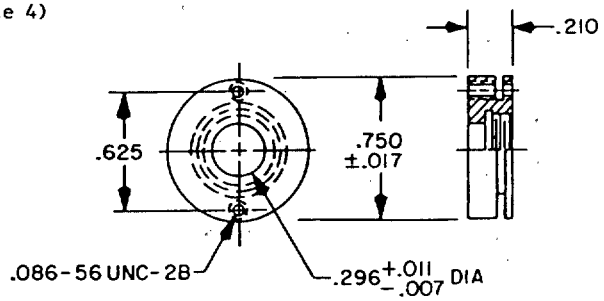



CONFIGURATION A

Dash number 1-01
(see note 4)




CONFIGURATION B

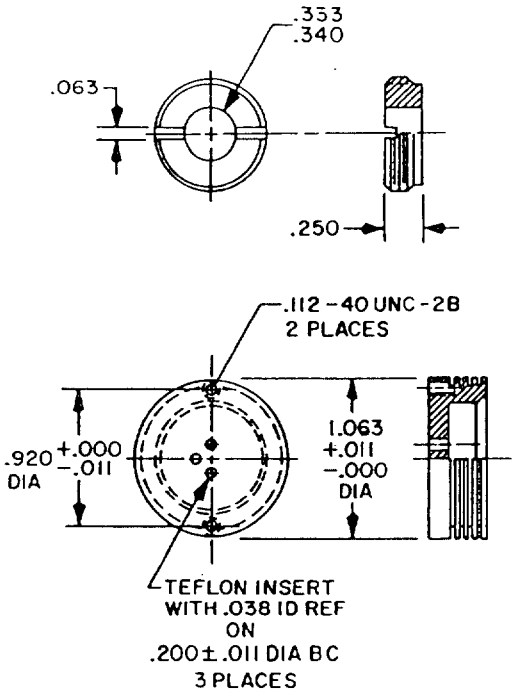
Dash number 1-02
(see note 4)

FIGURE 1. Heat sinks (encapsulating type).

(D) denotes changes

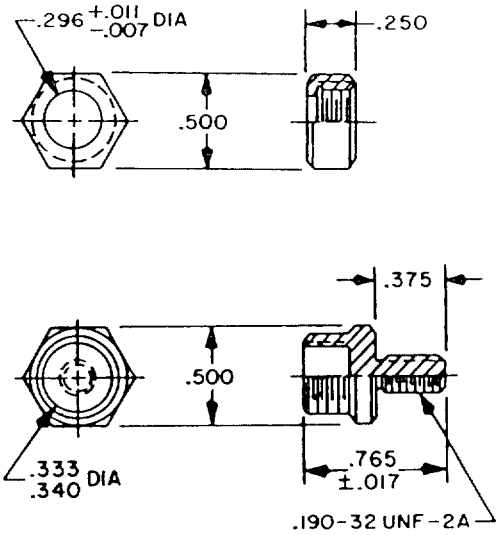

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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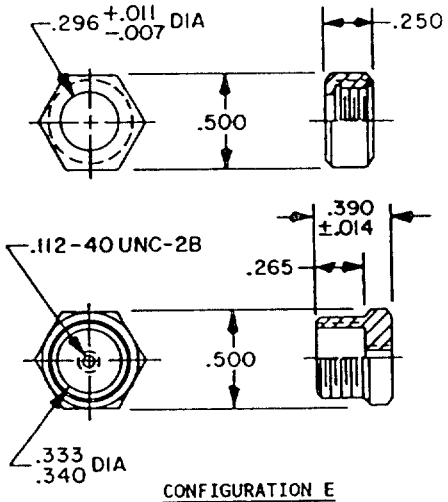

CONFIGURATION C

Dash number 1-03
(see note 4)



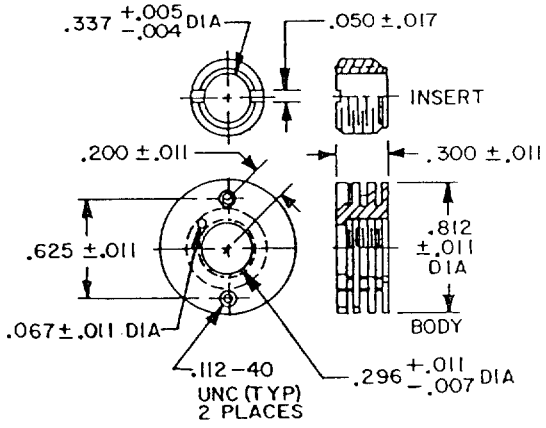

CONFIGURATION D

Dash number 1-04
(see note 4)




CONFIGURATION E

Dash number 1-05
(see note 4)



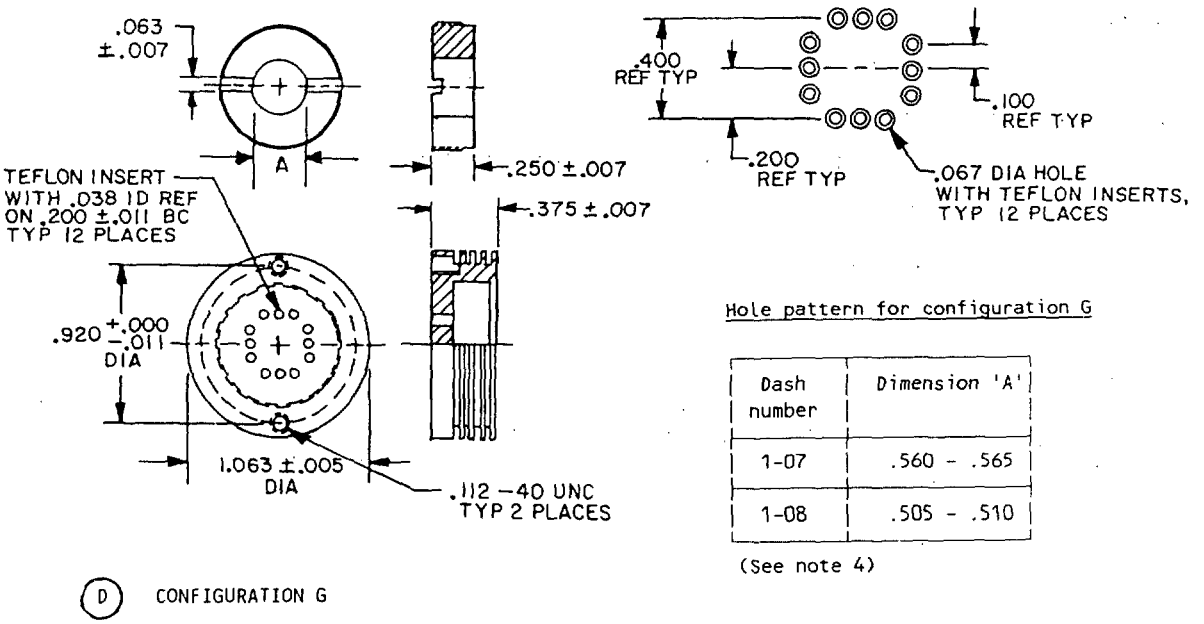

CONFIGURATION F

Dash number 1-06
(see note 4)

FIGURE 1. Heat sinks (encapsulating type) - Continued.



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Hole pattern for configuration G

| Dash number | Dimension 'A' |
|---|---|
| 1-07 | .560 - .565 |
| 1-08 | .505 - .510 |

(See note 4)

(D) CONFIGURATION G

| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .002 | 0.05 | .050 | 1.27 | .250 | 6.35 | .400 | 10.16 |
| .004 | 0.10 | .063 | 1.60 | .265 | 6.73 | .500 | 12.70 |
| .005 | 0.13 | .067 | 1.70 | .290 | 7.37 | .505 | 12.83 |
| .007 | 0.18 | .086 | 2.18 | .296 | 7.52 | .510 | 12.95 |
| .009 | 0.23 | .100 | 2.54 | .300 | 7.62 | .560 | 14.22 |
| .011 | 0.28 | .112 | 2.84 | .333 | 8.46 | .565 | 14.35 |
| .014 | 0.36 | .171 | 4.34 | .337 | 8.56 | .625 | 15.88 |
| .017 | 0.43 | .190 | 4.83 | .340 | 8.64 | .750 | 19.05 |
| .038 | 0.97 | .200 | 5.08 | .375 | 9.52 | .765 | 19.43 |
| | | .210 | 5.33 | .390 | 9.91 | .812 | 20.37 |
| | | | | | | .920 | 23.37 |
| | | | | | | 1.063 | 27.00 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.007 (0.18 mm). A .010 inch (0.25 mm) radius is permissible on corners and edges.
4. Complete PIN will include a finish letter in place of the hyphen shown (see finish requirement).

FIGURE 1. Heat sinks (encapsulating type) - Continued.



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REQUIREMENTS:

Design and dimensions: See figure 1.

Material: Aluminum alloy (class 1).

Finish: Black anodize (A), hard anodize (H), or insulating coating (C).

JEDEC case size: TO-5.

Thermal resistance: 60°C/watt, maximum.

Mounting surface flatness: Not applicable.

Surface roughness: Not applicable.

Part or Identifying Number (PIN). The term Part or Identifying Number replaces the term part number which was previously used in this specification sheet: M87111/04- (plus applicable dash number from figure 1 and finish letter as noted above). Complete PIN example: M87111/04-1H01.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - TD
Air Force - 19, 99
DLA - ES

User activities:
Navy - AS, CG, MC, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5999-0291)



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